First Progress Review

Chapter 3: Research Methodology

Calibrated Selective Autonomy in Multi-Agent Clinical Artificial Intelligence: A Clinician-Validated, Privacy-Preserving Framework for Brain-Tumour Diagnosis and Surgical-Planning Support in Low-Resource Settings

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3. Research Methodology

This chapter sets out how the research questions will be answered. It describes the research philosophy and design (Sections 3.1 and 3.2), the framework under study and the escalation rule that is its central mechanism (Sections 3.3 and 3.4), the methods for each objective and work package (Section 3.5), the populations, samples and data (Sections 3.6 and 3.7), hypotheses, measures and analysis (Sections 3.8 to 3.10), ethics (Section 3.11) and the technology stack and quality assurance (Section 3.12). The methodology follows the accepted proposal; choices of baselines, metrics and statistical tests have been refined using the methodology extracted from the Research Paper Bank (Chapter 2).

3.1 Research Philosophy and Approach

The research adopts a **pragmatist** philosophy: the value of the knowledge produced is judged by whether it works in its intended clinical setting, and methods are chosen for their fitness to each question rather than from a single paradigm (Creswell & Creswell, 2018; Saunders et al., 2019). The approach is predominantly **deductive**—hypotheses derived from the literature are tested on quantitative data—with an **inductive** strand in which clinicians' accounts of trust and escalation behaviour are analysed for themes. The strategy is **design-science research**, because the principal contribution is a constructed artefact, a working multi-agent clinical AI framework, evaluated against measurable criteria in its intended environment (Hevner et al., 2004).

3.2 Research Design

3.2.1 Design-science research process

The work follows the six activities of the design-science research methodology (Peffers et al., 2007). Table 3.1 maps each activity to this study. Design science is preferred over a purely experimental design because the central claim—that selective autonomy can be made calibrated, measurable and auditable—cannot be demonstrated by benchmarking a model; it requires building the mechanism into a working system and observing it with clinicians.

Table 3.1: Design-science research activities applied to this study.

Activity	In this study	Output
1. Problem identification	Literature review (Chapter 2); clinical pathway analysis with the clinical supervisor	Six gaps; problem statement
2. Objectives of a solution	Objectives O1-O4; hypotheses H1-H6; success criteria	Requirements and evaluation criteria
3. Design and development	Registry, agents, orchestrator, escalation policy, federated protocol (WP1-WP6)	Versioned artefact
4. Demonstration	Retrospective replay of registry cases through the framework	Audit-logged runs
5. Evaluation	Quantitative experiments (WP2-WP5) and multi-reader clinical evaluation (WP7)	Results against H1-H6
6. Communication	Papers, thesis, open benchmark and protocol	Publications, thesis

3.2.2 Mixed-methods evaluation

The evaluation is mixed-methods (Creswell & Creswell, 2018). The dominant **quantitative** strand comprises retrospective and simulated experiments on registry and public data, measuring segmentation, classification, calibration, escalation and federation performance, followed by a multi-reader, multi-case study. The supporting **qualitative** strand comprises semi-structured interviews and free-text feedback from clinicians on trust, usefulness and the acceptability of escalation behaviour. The strands are combined in an **explanatory sequential** arrangement: quantitative results identify where the framework helps or harms, and interviews explain why. Work proceeds in iterative build-and-evaluate cycles closed by clinician feedback; because the artefact evolves, every result is tied to a recorded model version, escalation-policy version, data version and random seed, and no result is carried forward across a change to either without re-evaluation.

3.3 The Framework Under Study

The framework is a team of specialised software agents coordinated by an orchestrator over a shared registry data layer (Figure 3.1). An **agent** is a self-contained component with one bounded job, a defined input and output, and one obligation that distinguishes it from an ordinary module: it must report a calibrated confidence in what it has produced. Six specialist agents cover the pathway: an **Intake Agent** (registration, de-duplication, de-identification and quality control), an **MRI Perception Agent** (detection, segmentation, volume and suggested type), a **Pathology Agent** (slide-level evidence for type and grade with attention maps), a **Diagnosis and Staging Agent** (fusion of imaging, tissue and available markers into a supported WHO 2021 diagnosis, grade and recommendation, degrading gracefully when a source is missing), a **Surgical Planning and Risk-Path Agent** (patient-specific 3D model, structures at risk and a candidate route with its anatomical uncertainty; proof of concept) and a **Follow-up Agent** (longitudinal comparison and surveillance scheduling).

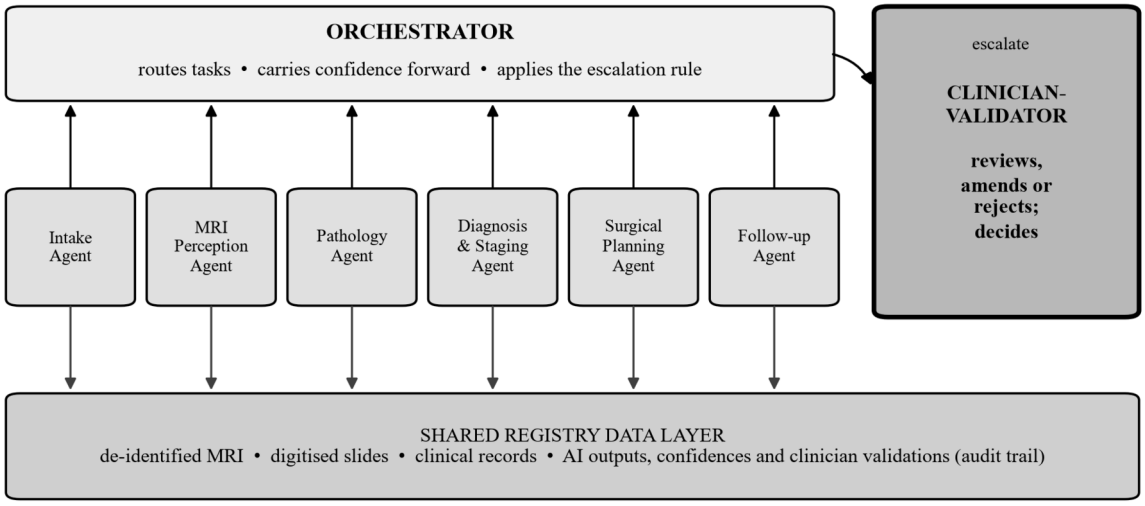


Figure 3.1: System architecture. Specialised agents are coordinated by an orchestrator over a shared registry data layer; every route to a clinical action passes the clinician-validator.

The **orchestrator** is not a diagnostic model. After every agent step it reads the agent's calibrated confidence, combines it with the clinical cost of error for that decision type, applies the escalation rule (Section 3.4) to choose one of three actions—*proceed*, *gather more evidence* or *escalate to a clinician*—and writes the input, output, confidence, action and reason to an append-only audit trail. The **clinician-validator** is a component with a defined role: escalated cases arrive with the agent's output, its evidence and the confidence

that triggered escalation, and the clinician accepts, amends or rejects it. Two invariants hold: there is no route from any agent to a clinical action of record that does not pass a clinician, and every escalation, autonomous step and clinician decision is logged. Figure 3.2 traces one case through the workflow.

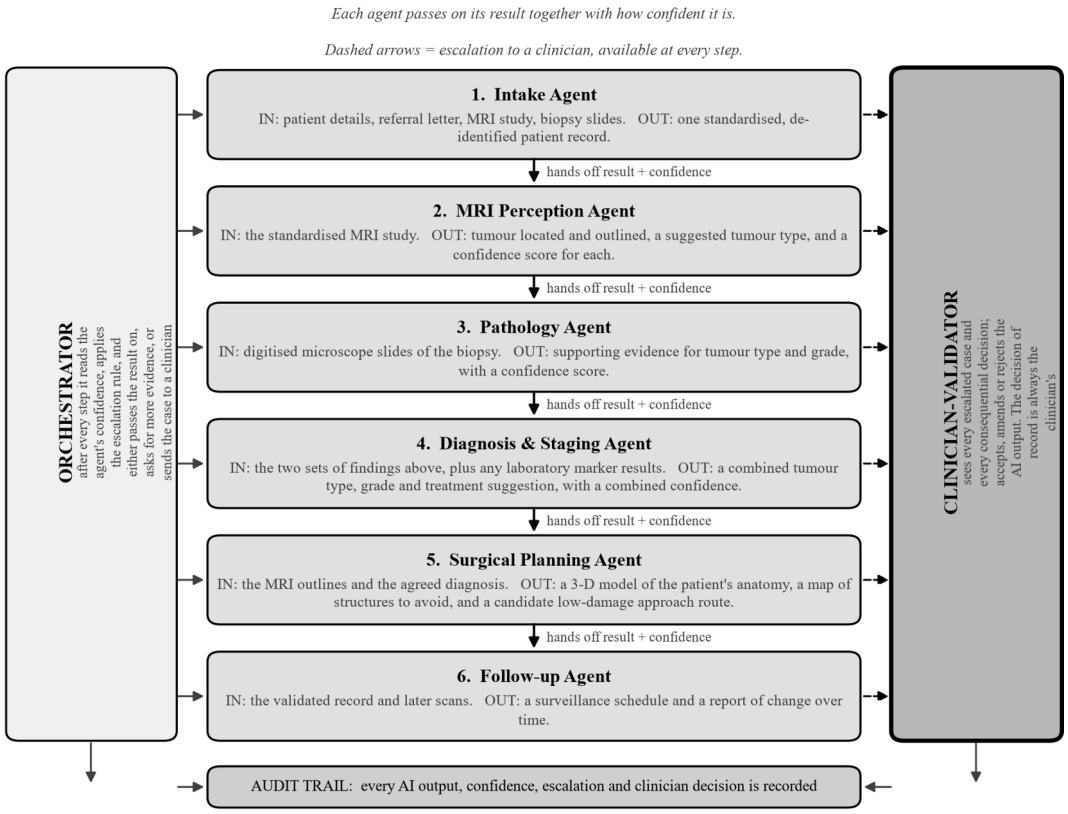


Figure 3.2: The agent workflow from input to final output. The orchestrator applies the escalation rule after every step, and escalation to a clinician is available at every step.

This design responds directly to the literature. Routing by measured, calibrated uncertainty rather than by language-model self-rating addresses the weakness of current medical multi-agent systems (Kim et al., 2024; Lou et al., 2026); isolating tasks in dedicated agents follows evidence that orchestrated designs sustain accuracy under load (Klang et al., 2025); and recording evidence for every claim responds to audits showing that unsupported observations propagate and consensus substitutes for evidence (Zhu et al., 2025). The specialist perception agents are conventional deep-learning models rather than language models, because combining specialist models with a coordinating layer and clinician intervention outperforms either generalist or specialist models alone in modality-specific analysis (Wang et al., 2026).

3.4 The Escalation Rule

The escalation rule is the mechanism under study (Figure 3.3). For agent k and case x, the raw model score is first calibrated so that a reported probability of 0.8 corresponds to being right about 80% of the time; candidate calibrators are temperature scaling (Guo et al., 2017), multi-domain temperature scaling (Yu et al., 2022) and ensemble averaging (Lakshminarayanan et al., 2017). The calibrated probability is converted into an uncertainty $u_k(x)$ (for classification, one minus the maximum calibrated probability or the predictive entropy; for segmentation, an aggregated voxel uncertainty, for which pairwise Dice between ensemble members is a strong baseline (Zenk et al., 2024)).

The threshold is derived from the clinical cost of the two kinds of error rather than chosen by hand. Let c_{FN} be the cost of wrongly reassuring (a missed or under-called finding), c_{FP} the cost of wrongly alarming, and c_R the cost of a clinician review, which reflects scarce specialist time. Acting autonomously on a prediction with calibrated probability p of being correct has expected cost (1 - p) times the cost of the relevant error; escalating costs c_R plus the residual clinician error. The agent therefore proceeds only when

$$(1 - p_k(x)) \cdot c_{err} \leq c_R + \varepsilon_{clin} \cdot c_{err} \quad (3.1)$$

where c_{err} is c_{FN} or c_{FP} according to the error the decision risks and ε_{clin} is the clinician's error rate on that decision. This gives a cost-derived threshold on calibrated confidence for each decision type. A middle band below the threshold triggers *gather more evidence* (for example, waiting for the pathology result before the diagnosis is released), and a list of decisions that are always escalated regardless of confidence (for example, any surgical recommendation, and any conflict between imaging and tissue evidence) is fixed in advance. Out-of-distribution inputs flagged by a separate detector are escalated (Yang et al., 2024). The costs are elicited from the clinical supervisor and participating clinicians through structured scenarios, not assumed, and the policy's sensitivity to them is reported.

Two further mechanisms address the chain of dependent decisions. **Uncertainty propagation:** each agent receives its upstream agents' uncertainty as an input feature and reports a combined uncertainty, so that a doubtful segmentation cannot become a confident grade. **Risk control:** the thresholds are calibrated on held-out data with conformal risk control (Angelopoulos et al., 2024) so that the error rate among autonomously handled cases stays below a target with a stated probability; because such guarantees are marginal and can fail under shift (Mehrtens et al., 2023), they are re-checked at each site and after

every model update. Baselines are a fixed confidence threshold on the uncalibrated softmax (the maximum softmax response, which remains a strong baseline (Bungert et al., 2023)), selective classification with guaranteed risk (Geifman & El-Yaniv, 2017), learning to defer with one-vs-all calibration (Verma & Nalisnick, 2022) and complementarity-driven deferral (Dvijotham et al., 2023).

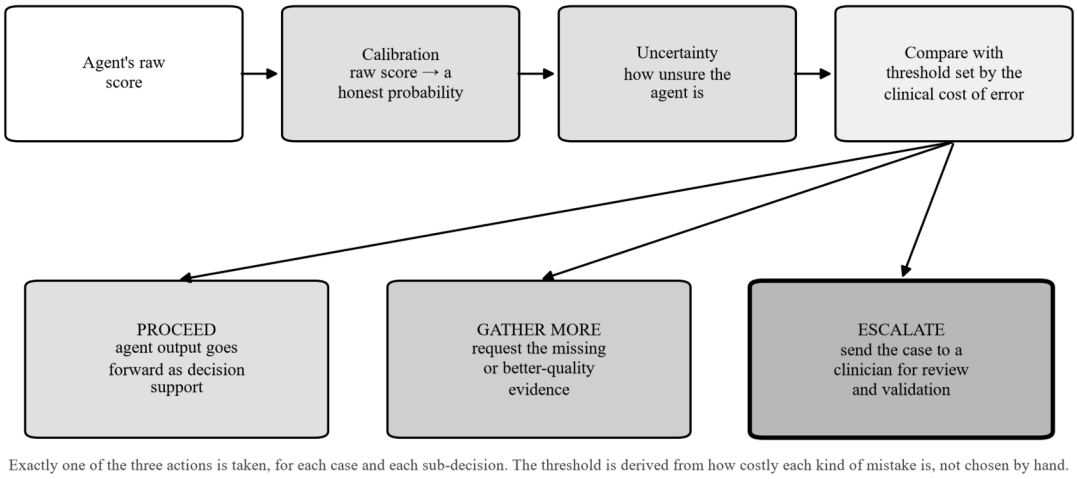


Figure 3.3: How the escalation decision is computed: a raw score is calibrated, converted into an uncertainty, and compared against a threshold derived from the clinical cost of error.

In variable terms, the **independent variables** are the design choices under the researcher's control: uncertainty-estimation method, calibration method, escalation policy and its cost parameters, training regime (local, transfer, self-supervised, federated) and fusion strategy. The **mediating variable** is the calibrated uncertainty attached to each agent output: the thesis claims that the effect of the design choices on outcomes is transmitted through the quality of this estimate rather than through raw accuracy. The **dependent variables** are diagnostic performance, clinician workload, escalation appropriateness, safety and trust. **Moderating variables** recorded for every case are scanner and protocol, image and slide quality, tumour type and grade, completeness of evidence and site.

3.5 Methods by Objective and Work Package

The work is organised into seven work packages (WPs). Table 3.2 maps them to objectives and research questions; the subsections give the methods.

Table 3.2: Work packages, objectives and research questions.

WP	Content	Objective	RQ
WP1	Registry platform and data pipeline (enabling)	O1	RQ1

WP	Content	Objective	RQ
WP2	Data-efficient MRI perception, calibrated uncertainty and local adaptation	O3	RQ3
WP3	Uncertainty-aware orchestration and escalation policy (scientific core)	O1, O2	RQ1, RQ2
WP4	Computational pathology and combined diagnosis and staging	O3	RQ3
WP5	Federated learning across hospitals	O3	RQ3
WP6	Surgical-planning agent (proof of concept)	O4	RQ4
WP7	Clinical validation of the human-in-the-loop framework	O4	RQ4

3.5.1 WP1: Registry platform and data pipeline (O1)

A web-based registry is built with a PostgreSQL database using an HL7 FHIR resource model (Bender & Sartipi, 2013) so that it interoperates with national systems, and an Orthanc server for DICOM imaging (Jodogne, 2018). The pipeline performs de-duplication, removal of direct identifiers, DICOM header scrubbing, defacing of structural MRI, removal of slide metadata, automated quality control of scans and slides, and quarantine of failed records. The audit trail records every agent output, orchestrator action and clinician decision with model and policy versions, following model-card and internal-audit practice (Mitchell et al., 2019; Raji et al., 2020).

3.5.2 WP2: MRI perception (O3)

Pre-processing uses N4 bias-field correction, registration to a common template, and HD-BET brain extraction validated on tumour MRI (Isensee et al., 2019). Segmentation baselines are nnU-Net (Isensee et al., 2021) and Swin UNETR (Hatamizadeh et al., 2022), with MedNeXt (Roy et al., 2023) as a further comparator; missing sequences are handled by modality-dropout training and a missing-modality model (Zhang et al., 2022). Classification of tumour type uses the segmented region and the full volume. Data efficiency is studied in three regimes: (a) training on public data only (BraTS 2021 (Baid et al., 2021)); (b) transfer learning with fine-tuning on local data (Boyd et al., 2024); and (c) self-supervised pre-training on unlabelled local scans followed by fine-tuning (Tang et al., 2022). Uncertainty is estimated with Monte Carlo dropout, deep ensembles and test-time augmentation, and calibrated as in Section 3.4. An ablation isolates the effect of self-supervised pre-training and adaptation, and a failure analysis assigns every case in the worst-performing decile to an error category.

3.5.3 WP3: Orchestration and escalation (O1, O2)

The orchestrator and escalation policy are implemented as in Sections 3.3 and 3.4 and evaluated by replaying retrospective registry cases (and public cases while local data accrue) through the full agent chain. Clinician decisions recorded in the registry act as the reference for what the clinician would have done. Experiments compare the policy against each baseline on risk-coverage curves, trace error propagation through the chain with and without uncertainty propagation (the direct test of H2), decompose escalations by trigger (low confidence, out-of-distribution, conflicting evidence, mandatory review), and vary the elicited costs to show the workload-safety frontier and its sensitivity.

3.5.4 WP4: Pathology and combined diagnosis (O3)

Biopsy slides are digitised, managed with OpenSlide and QuPath (Bankhead et al., 2017), stain-normalised, tiled and encoded with frozen pathology foundation-model encoders such as UNI (Chen et al., 2024). Slide-level type and grade are predicted with attention-based multiple-instance learning (CLAM) (Lu et al., 2021), giving attention maps for interpretability. The Diagnosis and Staging Agent fuses MRI and slide representations, and molecular markers where recorded, using late fusion and a mixture-of-experts design (Saueressig et al., 2026) trained with random source dropout so that it degrades gracefully when a source is missing. Concordance is reported for imaging alone, tissue alone and their combination, with an ablation under simulated missing sources.

3.5.5 WP5: Federated learning (O3)

Federated training is implemented with NVIDIA FLARE (Roth et al., 2022) across the participating hospitals, first simulated by partitioning data by site and then deployed. Aggregation methods compared are FedAvg (McMahan et al., 2017), FedProx (Li et al., 2020), SCAFFOLD (Karimireddy et al., 2020) and FedBN (Li et al., 2021); differential privacy is added where required (Abadi et al., 2016). Each federated model is compared with single-site and centrally trained models. Because a model that gains accuracy while losing calibration silently changes the deployed escalation behaviour, calibration error and the escalation policy's coverage and selective risk are re-measured after every aggregation round.

3.5.6 WP6: Surgical-planning proof of concept (O4)

Using the MRI segmentation and diffusion tractography, the agent builds a patient-specific 3D model in 3D Slicer (Fedorov et al., 2012), outlines eloquent cortex, major vessels and white-matter tracts, and computes a candidate approach route that minimises a risk-weighted cost with a sampling-based optimal planner (Karaman & Frazzoli, 2011). The uncertainty of the segmented structures is carried into the route as a risk margin. Evaluation is deliberately descriptive: geometric measures of planned routes and neurosurgeons' ratings of plausibility and usefulness.

3.5.7 WP7: Clinical validation (O4)

A multi-reader, multi-case study compares an unassisted workflow with the AI-assisted human-in-the-loop workflow. Each participating clinician (neurosurgeons, radiologists, histopathologists and senior trainees) reads a common set of cases under both conditions, with counterbalanced order and a washout period. Outcomes are diagnostic performance, review time, escalation appropriateness, usability, workload and trust (Section 3.9). Semi-structured interviews follow. The study is reported to DECIDE-AI (Vasey et al., 2022) and designed to detect the change in error pattern that selective prediction can induce in clinicians (Jabbour et al., 2025).

3.6 Population and Sample

Patients. The target population is all patients with primary or secondary brain and nervous-system tumours in Sri Lanka (about 700 new cases a year (International Agency for Research on Cancer, 2024)). The accessible population is patients presenting to the participating tertiary units during the study period, recruited by consecutive sampling. Eligible patients are of any age with a suspected or confirmed brain or nervous-system tumour, at least one usable MRI examination in DICOM form, sufficient clinical information to establish the reference standard, and ethics approval with consent or waiver. Records are excluded where imaging is unusable, the reference standard cannot be established, de-identification cannot be completed, or governance requirements are not met. Patients are not excluded for missing tissue, marker or follow-up data; they contribute according to data availability, which allows evaluation under realistic incompleteness. The expected cohort is several hundred patients over the study period.

Sample-size reasoning. For model development, the required sample is estimated with the Riley criteria for prediction models (Riley et al., 2020); for diagnostic accuracy, from the expected sensitivity and the width of its confidence interval (Obuchowski, 1998). As an illustration, estimating a sensitivity of 0.90 with a 95% confidence interval of ±0.07 requires about 71 positive cases for each class of interest. Because local numbers are bounded by incidence, public data are used for pre-training and baselines, and federated training increases the effective sample without pooling data.

Clinicians. Clinicians are recruited by purposive sampling based on expertise. The multi-reader, multi-case design uses the scarce readers efficiently: with about 6-8 readers and 60-100 cases, each read under both conditions, the study is powered to detect a clinically meaningful reduction in review time and to test non-inferiority of diagnostic accuracy within a pre-specified margin. **Public data** (BraTS 2021, BraTS-Africa, TCGA glioma slides and other open sets) are used for pre-training, benchmarking and external comparison only, never as the sole basis for a clinical claim.

3.7 Data Sources and Collection

Primary data are diagnostic MRI studies in DICOM, digitised biopsy slides, structured clinical information, operative and pathology reports and follow-up records. Secondary data are public benchmarks and national statistics (International Agency for Research on Cancer, 2024; National Cancer Control Programme, 2024). Reference annotations are produced with 3D Slicer (Fedorov et al., 2012) and QuPath (Bankhead et al., 2017) by trained annotators and reviewed by clinicians; a subset is double-annotated to measure inter-rater agreement. Diagnostic labels come from clinician-confirmed records using WHO 2021 terminology (Louis et al., 2021). Collection starts only after ethics approval; in federated experiments only model parameters leave a hospital (Figure 3.4).

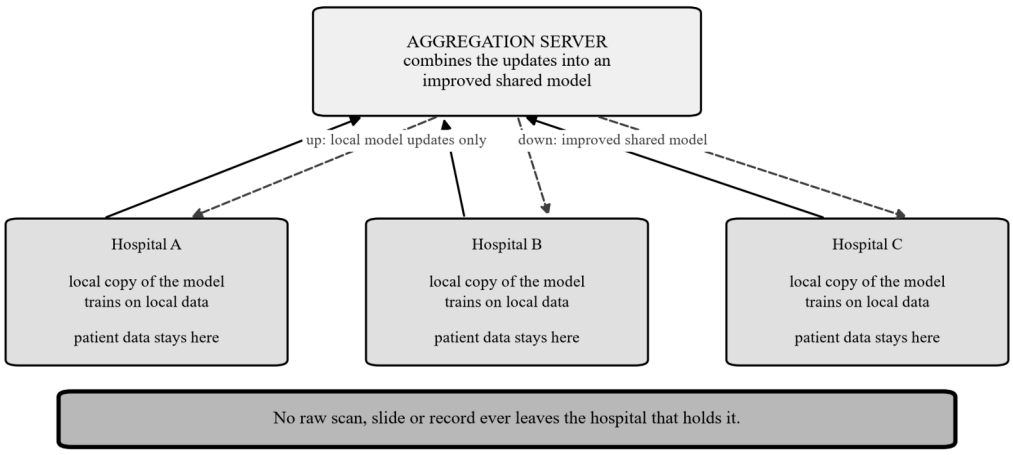


Figure 3.4: Federated training across participating hospitals. Only model parameters are exchanged; raw patient data never leave an institution.

3.8 Hypotheses

The hypotheses are stated with their null form of no difference between compared methods or conditions:

- **H1 (RQ2).** A calibrated, cost-derived escalation policy achieves a better workload-safety trade-off than fixed-confidence thresholding: lower error at the same escalation rate, or lower escalation at the same error rate.
- **H2 (RQ1).** Propagating calibrated uncertainty between agents reduces downstream decision errors compared with hand-offs that ignore upstream reliability.
- **H3 (RQ3).** Self-supervised pre-training on unlabelled local MRI followed by fine-tuning improves performance and calibration compared with transfer from public data alone.
- **H4 (RQ3).** Combining MRI and slide evidence gives higher diagnostic and grading concordance with the reference standard than the best single source.
- **H5 (RQ3).** Federated training improves generalisation over single-site training while staying within a pre-defined margin of centralised training and preserving calibration within a pre-defined tolerance.
- **H6 (RQ4).** The AI-assisted human-in-the-loop workflow reduces clinician review time and improves usability and trust compared with an unassisted workflow, while keeping diagnostic performance within a pre-defined non-inferiority margin.

3.9 Measures

Table 3.3: Operationalisation of the main constructs.

Construct	Measures
Segmentation quality	Dice similarity coefficient, 95th-percentile Hausdorff distance, normalised surface distance, sensitivity and specificity, per case with confidence intervals (Maier-Hein et al., 2024)
Classification and diagnosis	AUROC, balanced accuracy, per-class F1, concordance with the clinician reference standard
Calibration (mediator)	Expected and maximum calibration error (Naeini et al., 2015), Brier score (Brier, 1950), reliability diagrams (Guo et al., 2017), empirical coverage of conformal sets (Angelopoulos & Bates, 2021)
Selective autonomy	Coverage (share handled autonomously), selective risk (error among those), area under the risk-coverage curve (Geifman & El-Yaniv, 2017), appropriate-escalation rate (escalated cases in which the clinician changed the output), missed-escalation rate (autonomous cases judged by blinded experts to need a clinician; primary safety indicator)
Workload	Review minutes per case from the audit log; NASA Task Load Index (Hart & Staveland, 1988)
Usability and trust	System Usability Scale (Brooke, 1996); Likert trust instrument; interview themes
Agreement	Cohen's kappa (Landis & Koch, 1977) and Bland-Altman analysis for continuous measures (Bland & Altman, 1986)
Federation	Federated-minus-centralised performance, calibration change after aggregation, communication cost, privacy budget, audit-verified absence of raw-data transfer

3.10 Data Analysis

Models are developed with patient-level stratified cross-validation and reported once on a held-out local test set untouched during development. Splits are stratified by tumour type and site so that no patient appears in both training and test data; leave-one-site-out evaluation is added where generalisation across sites is the question. For H1 and H2, risk-coverage curves and their areas are compared with paired bootstrap confidence intervals from resampling patients (Efron & Tibshirani, 1993), and the missed-escalation rate at matched coverage is compared with McNemar's test (McNemar, 1947). For H3-H5, paired predictions on the same cases are compared with DeLong's test for AUROC (DeLong et al., 1988), McNemar's test for accuracy and the Wilcoxon signed-rank test for per-case metrics such as Dice. H6 uses mixed-effects models with reader and case as crossed random effects. Family-wise error is controlled within each work package by Holm's procedure (Holm, 1979), with primary endpoints declared in advance. Interviews are analysed by reflexive thematic analysis (Braun & Clarke, 2006) and triangulated with the escalation data. Negative results are reported in the same detail as positive ones, and diagnostic and prediction studies are reported to STARD-AI and TRIPOD+AI (Sounderajah et al., 2025; Collins et al., 2024).

3.11 Ethical, Privacy and Governance Considerations

Ethics approval is obtained before any patient data are accessed. The application to the Research Ethics Committee, Faculty of Medicine, KDU, was finalised with a KDU co-investigator and its processing fee paid on 10 November 2026 for submission; the University of Sri Jayewardenepura application follows KDU approval, and approval from each participating hospital is obtained before its data are used. De-identification is applied at ingestion, access is role-based and audited, and federated training keeps raw data within each hospital. The system is a decision-support tool: it issues no unvalidated clinical decision, defers under uncertainty, and never performs, controls or guides a surgical procedure; all clinical decisions remain the responsibility of the treating clinician (World Health Organization, 2021; European Union, 2024). Fairness is checked by reporting performance by site, age group, sex and tumour type, because bias can arise from apparently reasonable design choices (Obermeyer et al., 2019).

3.12 Technology Stack and Quality Assurance

Models are developed in Python with PyTorch and MONAI (Cardoso et al., 2022); segmentation baselines use nnU-Net and Swin UNETR; pathology uses OpenSlide, QuPath and CLAM with foundation-model encoders; federated learning uses NVIDIA FLARE with Flower as an alternative; surgical planning uses 3D Slicer and the Open Motion Planning Library; the registry uses PostgreSQL, Orthanc and HL7 FHIR. Reproducibility is ensured by experiment tracking, dataset versioning, containerised environments and fixed seeds, so that every reported figure can be regenerated from a recorded model, data and policy version. Code is kept under version control, and progress is recorded on the supervisors' progress website.

3.13 Validity and Reliability

- **Internal validity:** patient-level splits, a held-out test set, pre-declared endpoints and blinded expert review of autonomous cases guard against optimistic estimates.
- **External validity:** leave-one-site-out testing, public external sets and the federated design test generalisation across scanners and institutions.
- **Construct validity:** escalation quality is measured directly (appropriate and missed escalations) rather than inferred from accuracy.

- **Reliability:** double annotation with agreement statistics, repeated runs with different seeds, and versioned artefacts.
- **Qualitative trustworthiness:** an audit trail of coding decisions, triangulation with quantitative escalation data, and member checking with participating clinicians.

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